NEP Haryana Excellence Awards 2026: Revised VM Sizing and Calculations

Prepared	2026-10-21 (backup, recovery and load-test plan revised 2026-10-22)
Replaces	The sizing in `VM_SERVER_REQUIREMENTS.md` §3–§4 and `SERVER_REQUIREMENTS_SUMMARY.md` §2 (2026-10-18/19). Those figures were made without any user or volume data.
New inputs	At most **250–300 concurrent users**. About **40 applications last year**, so this year is planned for **at most 150 applications**. Upload limits (decided 2026-10-22): **PDFs 5 MB**, other evidence formats (images, DOCX, XLSX) **25 MB**.

Labels used below: **[VERIFIED]** means read from the code or the demo database. **[ASSUMPTION]** means an input that the calculation depends on, which HSHEC should confirm. **[ESTIMATE]** means a typical figure for this software stack, to be confirmed by the load test (§10).

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1. Result

#	Role	vCPU	RAM	Storage	Change from the earlier proposal
1	**Production**: Nginx, Gunicorn/Django, PostgreSQL 16 and Redis on one VM	**4**	**8 GB**	**60 GB** SSD (OS, app, database) + **100 GB** SSD separate, expandable evidence volume	Replaces two VMs (App 4/8 + DB 4/8). The separate database VM is dropped.
2	**Staging / UAT**: all components	**2**	**4 GB**	**60 GB** SSD	was 2 / 4–8 GB / 100 GB
3	**Backup storage**, off the production VM	–	–	**250 GB**	was 1 TB
	Total	**6**	**12 GB**	**220 GB SSD + 250 GB backup**	was 10 vCPU, 20–24 GB, ~550 GB SSD + 1 TB backup

That is **40% fewer vCPU, 40–50% less RAM, 60% less SSD and 75% less backup storage**. All servers run Ubuntu Server 24.04 LTS, and the software list is unchanged.

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2. Inputs

#	Input	Value used	Source
I1	Applications	**150** (last year: 40)	HSHEC, 2026-10-21
I2	Peak concurrent logged-in users	**300**	HSHEC, 2026-10-21 (upper end of 250–300)
I3	Evidence files per complete application	**50**	**[VERIFIED]** The complete demo applications hold 46 (college) and 48 (university) evidence documents. Rounded up to 50.
I4	Average evidence file size	**3 MB**	**[ASSUMPTION]** Scanned PDFs are typically 1–5 MB. PDFs are limited to **5 MB** (`MAX_PDF_FILE_SIZE`) and other formats to **25 MB** (`MAX_EVIDENCE_FILE_SIZE`) **[VERIFIED]**, enforced in the browser and on the server. 146 of the 151 evidence slots also accept images or Office files **[VERIFIED]** (evidence taxonomy).
I5	Re-upload / version headroom	**× 1.5**	**[ASSUMPTION]** Replaced files are kept as superseded versions **[VERIFIED]** (`EvidenceDocument.superseded_by`), and rejected files go to `quarantine/`.
I6	Time between requests for one active user	**10 s** at peak (20 s normal)	**[ASSUMPTION]**, conservative for form filling
I7	API response time / CPU time per request	**200 ms / 50 ms**	**[ESTIMATE]** for Django REST Framework with PostgreSQL on the same host
I8	Memory per Gunicorn worker process	**150 MB**	**[ESTIMATE]** for Django 6 + DRF + pypdf
I9	Submission peak	**40% of applications on the final day**, over 8 working hours, with the busiest hour at **3×** the average	**[ASSUMPTION]**, typical deadline behaviour
I10	Backup retention	**35 days**	Existing operations checklist (to be confirmed, Q8)

3. Request load

```
Peak request rate   = concurrent users ÷ time between requests
                    = 300 ÷ 10 s                = 30 requests/second
Normal request rate = 300 ÷ 20 s                = 15 requests/second

Evidence uploads on the final day = 150 × 40% × 50 files     = 3,000 files
Average upload rate               = 3,000 ÷ (8 h × 3,600 s)   = 0.10 files/second
Busiest-hour upload rate          = 0.10 × 3                  = 0.31 files/second
```

30 requests per second means every one of the 300 users clicks or saves every 10 seconds at the same time. This is a deliberately high figure.

4. CPU

```
Django CPU      = 30 req/s × 0.05 s                        = 1.50 cores
PostgreSQL CPU  ≈ 40% of Django CPU                        = 0.60 cores
Upload checks   = 0.31 files/s × 0.5 s (PDF parse, hash)   = 0.16 cores
Nginx + Redis + OS                                         ≈ 0.25 cores
                                                           -------------
Peak demand                                                ≈ 2.5 cores
```

VM size	Peak utilisation	Assessment
2 vCPU	2.5 ÷ 2 = **125%**	Overloaded at the deadline peak. Requests queue and slow down.
4 vCPU	2.5 ÷ 4 = **63%**	**Within the 70% planning limit.** Normal load (15 req/s) is about 1.3 cores, or 33%.
8 vCPU (earlier proposal, single-VM option)	31%	More than needed

Result: 4 vCPU.

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5. Concurrency (Gunicorn workers)

By Little's law, the number of requests in progress equals the arrival rate multiplied by the response time:

```
API requests in progress = 30 req/s × 0.2 s                      = 6
Uploads in progress      = 0.31 files/s × 2 s (incl. PDF parse)  = 0.6 → allow a burst of 5
                                                                 ---
Request slots needed                                             ≈ 11
```

Configuration: **5 worker processes × 3 threads = 15 slots** (`gunicorn --workers 5 --threads 3`). This leaves about 35% spare. Nginx receives the whole upload before passing it to Gunicorn, so a user's slow connection does not hold a worker.

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6. Memory

Component	Calculation	RAM
Operating system, monitoring agents		1.00 GB
Gunicorn workers	5 × 150 MB	0.75 GB
Files held in memory (worst case)	15 slots × 25 MB × 2 (parsing overhead). Downloads are read fully into memory **[VERIFIED]**.	0.75 GB
PostgreSQL	shared_buffers 512 MB + 20 connections × 4 MB work_mem + 128 MB maintenance	0.75 GB
Redis	lockout, throttle counters and a user cache for 300 users	0.10 GB
Nginx		0.05 GB
Total at peak		**≈ 3.4 GB**

VM size	Peak use	Assessment
4 GB	**85%**	Too little room. A worst-case burst of large uploads could cause swapping or an out-of-memory kill.
8 GB	**43%**	**Recommended.** The remainder is used as disk cache, which keeps the whole database and recently used evidence in memory. A 4 vCPU VM normally comes with 8 GB anyway.

Result: 8 GB.

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7. Storage

7.1 Evidence volume

```
Evidence = applications × files per application × average size × version headroom
         = 150 × 50 × 3 MB × 1.5
         = 33,750 MB  ≈ 34 GB
```

Scenario	Calculation	Size
Last year's volume (40 applications)	40 × 50 × 3 MB × 1.5	9 GB
Planning case	150 × 50 × 3 MB × 1.5	**34 GB**
Larger files (5 MB average)	150 × 50 × 5 MB × 1.5	56 GB
Theoretical ceiling (every file at its limit, no re-uploads)	150 × (49 × 25 MB + 1 × 5 MB)	185 GB (not realistic)

Because almost every slot accepts 25 MB image or Office files, the limits do not cap storage at a small figure. **Allocate 100 GB** on an expandable volume (LVM or cloud disk), with an alert at 70%. 100 GB holds the planning case three times over and still fits an average file size of up to about 9 MB. The earlier 250 GB was based on all 1,152 institutions applying.

7.2 OS / application / database disk

Item	Size
Ubuntu, packages, Python virtual environment, built frontend	15 GB
PostgreSQL data + WAL. The demo database is 2.5 MB for 3,769 rows including 8 complete demo applications **[VERIFIED]**. Even at 5 MB per application: 150 × 5 MB = 0.75 GB, doubled for indexes and WAL.	2 GB
Local copy of the latest backup before it is shipped off the VM	5 GB
Logs, 30 days compressed (≈ 160 MB/day uncompressed at 15 req/s for 10 h)	2 GB
Temporary space for uploads larger than 2.5 MB **[VERIFIED]**	5 GB
Free space for upgrades, at least 50%	31 GB
Total	**60 GB**

7.3 Backup storage

Evidence is backed up **daily**, not weekly. Weekly copies would lose up to a week of uploads if the disk failed near the deadline. Daily full copies would need 35 × 34 GB ≈ 1.2 TB, so the evidence uses a deduplicating tool (restic,

borg or storage snapshots). That stores each file once, plus about 20% for metadata and deleted files kept within retention.

```
Database: daily compressed dump ≤ 0.5 GB × 35 days, plus WAL archive   ≈ 20 GB
Evidence: daily deduplicated backups, 34 GB × 1.2                     ≈ 41 GB
                                                                      --------
Planning case                                                         ≈ 61 GB
Evidence volume full (100 GB × 1.2 + 20 GB)                           ≈ 140 GB
```

250 GB covers a full evidence volume with about 45% to spare.

Recovery target	Value	How
Database data loss (RPO)	≤ 1 hour	Continuous WAL archiving to backup storage
Evidence data loss (RPO)	≤ 24 hours	Daily deduplicated backup
Recovery time (RTO)	≤ 4 hours	Rebuild on a new VM from backup; practised in a restore drill on staging before go-live
Before the window opens and before each deployment	VM snapshot	Fast roll-back for a bad deployment

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8. Bandwidth

Traffic in the busiest hour	Calculation	Rate
Evidence uploads	0.31 files/s × 3 MB × 8 bits	7.4 Mbps
Evidence downloads by reviewers (assumed equal to uploads)		7.4 Mbps
First page load (2.7 MB application, before compression) for all 300 users in one hour	300 × 2.7 MB × 8 ÷ 3,600 s	1.8 Mbps
API traffic	30 req/s × 20 KB × 8 bits	4.8 Mbps
Total		**≈ 21 Mbps**

At least 50 Mbps dedicated, or 100 Mbps shared. At 50 Mbps, a 5 MB PDF takes under 1 second of line time and a 25 MB file about 4 seconds. In practice, the user's own connection is usually the limit.

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9. Why one production VM is enough

- **The database is small.** About 1–2 GB at the end of the cycle, so it sits entirely in memory on the same VM. A second VM adds a network hop, a second server to patch and a second firewall rule set, but no extra capacity is needed.
- **Availability is unchanged.** The earlier two-VM design also had a single application server and a single database server. Either one failing stopped the service. Recovery in both designs relies on the backups and recovery targets in §7.3 (at most 24 hours of evidence lost, service back within 4 hours).
- **Scaling is simple if needed.** The VM can be resized to 6–8 vCPU / 12–16 GB without code changes. PostgreSQL can move to its own VM or a managed service later by changing `DATABASE_URL`.
- **Two separate disks are still kept.** Evidence lives on its own volume, so it can grow and be backed up separately from the OS disk.

The VM must be allowed to resize quickly (a reboot within an hour) if the load test or the first week of live use shows CPU or memory above 70%.

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10. Confirming these figures

The **[ESTIMATE]** figures (I7, I8) are typical values, not measurements from this application. Staging has half the CPU and memory of production, so a 300-user test there would fail whatever production can do. Run the tests in two steps:

1. **Staging, 150 users:** a trial run to find errors and tune the test script.
2. **Production, 300 users:** the real test, run before go-live while the VM holds no real data. Test accounts and files are removed afterwards.

Targets for the production test:

Item	Target
Virtual users	300, each acting every 10 s: login, open the assessment, save, upload a 3–5 MB PDF and a 25 MB image, download evidence
Response time	95% of requests under 1 s; uploads under 5 s
Error rate	Under 1%
CPU / memory	Under 70% on the production VM
Cloudinary uploads	Uploads to Cloudinary must succeed through the data-centre firewall or proxy. Upload time counts toward the 5 s target.

One figure from HSHEC would replace the main assumption: the total size of last year's evidence for the 40 applications. Dividing it by 40 gives the actual size per application, which replaces the estimate of 50 × 3 MB × 1.5 = 225 MB.

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11. Not changed

The software stack, network ports, domain/SSL/email requirements, security baseline and open points (Q3–Q13) in `VM_SERVER_REQUIREMENTS.md` still apply, with two changes:

- On the single VM, PostgreSQL listens on localhost only, so the internal port 5432 rule between VMs is no longer needed.
- Outbound HTTPS from VM 1 and VM 2 to `api.cloudinary.com` is required. The nomination evidence upload sends files from the server to Cloudinary [VERIFIED] (`Backend/apps/nominations/views.py`). Approval for third-party storage is open question Q5.
- The request size allowed by Nginx and any firewall or WAF stays at **30 MB** (`client_max_body_size 30M`: a 25 MB file plus form overhead).